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Microstructural evolution and mechanical properties of alumina ceramics joints fabricated by ultrafast laser welding

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Highlights

- The sound welded sample of alumina ceramic is achieved by the direct ultrafast laser.
- The weld seam presents the typical columnar dendrites, composed of predominantly α-Al_2O_3 and minor glassy phase.
- The shear strength of the welded sample is 83 MPa, higher than that of traditional brazed joints.

Abstract

Outline

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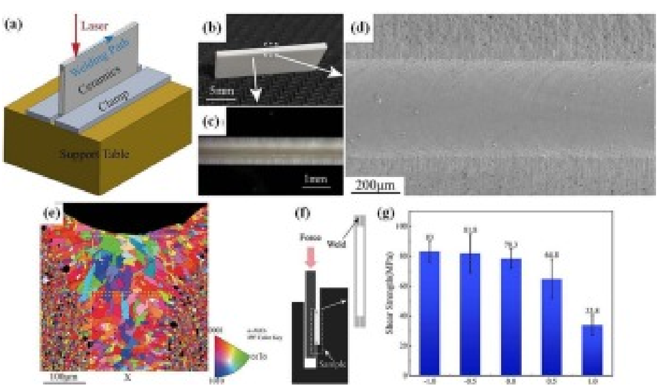
Ultrafast laser welding presents a viable method for achieving the direct joining of alumina ceramics. Nevertheless, current investigation on the ultrafast laser welding of alumina ceramics remains limited. The work comprehensively discusses the effects of three critical process parameters (defocus distance (f), single-pulse energy (E), and welding speed (v)) from ultrafast laser welding on the weld formation, microstructure evolution and mechanical properties of the joint. Experimental results demonstrate that the defect-free joints with superior quality are achieved through the optimized parameters: defocus distance of −1 mm, single pulse energy of 17.6 μJ, and welding speed of 1 mm/s. The weld seam is characterized by the predominant α-Al_2O_3 phase and few intergranular glass phase (SiO_2), exhibiting the highest shear strength, about 90.9 MPa (comparable to the conventional brazing/diffusion methods). In addition, the above method enhances the joining efficiency of alumina ceramics. This work provides the strong evidence for the feasibility of high-quality fusion welding of alumina ceramics by ultrafast laser welding.

Graphical abstract



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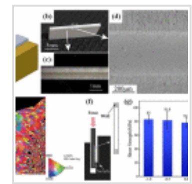
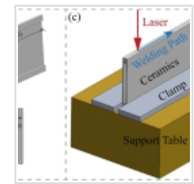
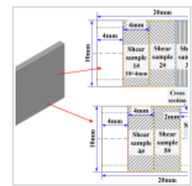
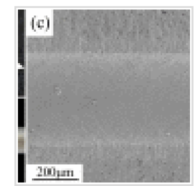
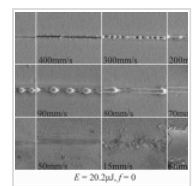
Figures (21)

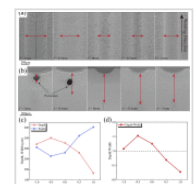












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Tables (1)

Table 1. The welding parameters of ultra...

Ultrafast laser welding; Alumina ceramics; Microstructural evolution; Welding defects; Shear strength

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1. Introduction

Alumina ceramics are the most widely utilized engineering ceramic material, with applications extending across biomedical, precision machining, chemical engineering, electronic packaging, and aerospace fields [1,2]. In the practical implementations, reliable joining techniques are essential for constructing complex ceramic components. However, alumina ceramics possess inherent characteristics (such as high melting point, significant brittleness, poor thermal shock resistance, inadequate wettability, electrical non-conductivity, et al.), which fundamentally limit the application of conventional welding methods in joining alumina ceramics. Conventional welding methods (e.g., brazing [1], diffusion welding [4], high-temperature bonding [5], continuous laser welding [6,7]) generally demand that entire ceramic components undergo both high-temperature heating and prolonged heat preservation, which leads to higher energy consumption and lower production efficiency. In addition, in the field of electronic packaging, electronic components on ceramic substrates are often difficult to withstand high temperatures when heated as a whole. Therefore, modern manufacturing requires a new technology for joining alumina ceramics. Furthermore, in electronic packaging applications, electronic components mounted on ceramic substrates are often susceptible to damage during bulk heating processes due to their limited tolerance for high temperatures. The inherent constraint necessitates the development of novel joining techniques for alumina ceramics that are compatible with modern manufacturing requirements.

Ultrafast lasers, characterized by pulse durations on the femtosecond to picosecond timescale, are distinguished by their extremely short pulse widths and ultra-high peak power densities [8,9]. Systems delivering microjoule-level pulse energies can achieve instantaneous power outputs reaching the terawatt (TW) scale. When interacting with ceramic materials, these high-intensity pulses induce nonlinear absorption, material removal, and melting within an ultrashort temporal window. Unlike continuous-wave lasers, ultrafast lasers enable precise energy deposition confining thermal effects to a highly localized region. The mechanism significantly minimizes the heat-affected zone (HAZ) by curtailing thermal conduction [10] and reduces associated thermal stresses. Consequently, ultrafast laser processing offers distinct advantages for materials with high melting points and brittleness, presenting considerable potential for advanced ceramic welding applications [11]. At present, the relevant research mainly focuses on the joining of glass materials [12,13], and has been excellent welding quality. Penilla et al. achieved the joining of yttria-stabilized zirconia (YSZ) by ultrafast laser welding. The joint showed the highest shear strengths of 40 MPa, which were comparable to joints obtained using diffusion bonding of ceramics to metals [11]. Li et al. discussed the influence of femtosecond laser welding process parameters on the quality of alumina joint. They found that the influence degree of welding process parameters on weld penetration was welding speed, laser power and defocusing distance in turns [14]. Yuan et al. have been conducted a series of studies on ultrashort pulse laser welding of ceramics [[15], [16], [17]]. They obtained the defect-free ceramic joint by ultrashort pulse laser and focused on exploring the effect of welding process parameters on the microstructure and mechanical properties of joints. But the solidification process of molten pool and welding defects in the joints are limited.

The work investigates the melting behavior of alumina ceramics induced by ultrafast laser irradiation. The effects of key process parameters—including defocus distance, single-pulse energy, and welding speed—on the macroscopic morphology, microstructure, defect formation, and mechanical properties of the welds are systematically examined. Furthermore, the underlying mechanisms governing these effects are elucidated.

2. Materials and methods

The α-alumina ceramics used in this work were manufactured by Baike Electronics, with a purity of 95–96 wt% α-Al_2O_3. The α-alumina ceramics has a density of 3.75 g/cm^2, and contains 4–5 wt% glassy phase impurities. The glassy phase impurities are caused by the addition of silicon dioxide (SiO_2), small amounts of magnesium oxide (MgO), and calcium oxide (CaO) during sintering.

The dimensions of the alumina ceramic components are shown in Fig. 1a and b. Monaco 1035-40-40 ultrafast laser manufactured by Coherent, USA was applied in joining α-alumina ceramics. The system generates Gaussian pulses with the following specifications: 1035 ± 5 nm wavelength, 300 fs–10 ps pulse width (full-width at half-maximum), 1 MHz pulse repetition rate, and spatially Gaussian-distributed energy density profiles. After transmitted through the optical path, the waist diameter of the beam is measured to be 20 μm, and the corresponding Rayleigh length is approximately 303.5 μm. Prior to welding, the alumina ceramic specimens were subjected to a two-stage pretreatment. First, they were ultrasonically cleaned in anhydrous ethanol for 3 min to remove surface contaminants. Subsequently, the specimens were placed on a temperature-controlled hotplate and held at 115 °C for 60 min to ensure complete evaporation of residual alcohol and adsorbed moisture. The schematic diagram of welding clamp is illustrated in Fig. 1c. Two-piece ceramics are stacked together, and then welded along the edges. In order to suppress the thermal shock cracks, the laser beam will swing 5 times to both sides with an amplitude of 0.2 mm within the range of about 0.5 mm in length of the welding starting position, playing preheating role.

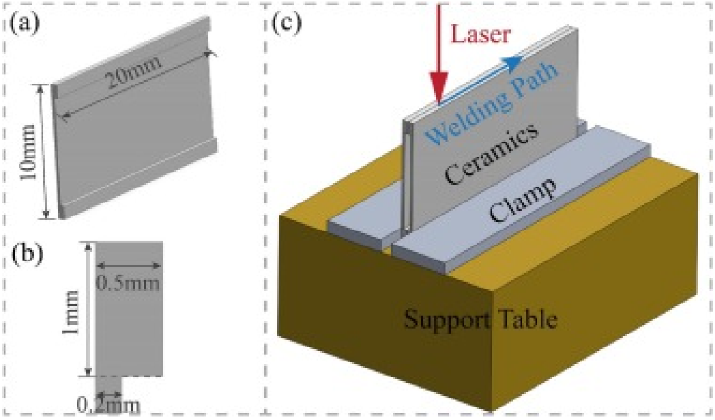

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Fig. 1. (a) Dimensions of the alumina ceramic components; (b) clamping and welding methods.

The sampling location of metallographic specimens are shown in Fig. 2. Metallographic analysis was performed using the ZEISS Axio Observer 3 inverted metallurgical microscope. Macroscopic topography observation was performed by the ZEISS Stemi 508 stereomicroscope system. SEM (Scanning Electron Microscope) images were acquired using the Thermo Fisher Scientific Quattro ESEM environmental scanning electron microscope. TEM (Transmission Electron Microscopy) images and EDS (Energy Dispersive Spectroscopy) were acquired using the Thermo Fisher Scientific Talos F200X G2 transmission electron microscope. EBSD (Electron Backscatter Diffraction) images were acquired using the Oxford Instruments Nordlys Max3 EBSD detector system. XRD (X-ray Diffraction) analysis was performed using the Bruker D8 Advance X-ray diffractometer. Conduct hardness tests using HVS-1000Z digital microhardness tester and shear strength tests using UTM5105-G electromechanical universal testing machine. In order to ensure the accuracy of the shear properties of the joint, 5 specimens are cut according to the position shown in Fig. 2 and tested. The final shear strength is taken as the average of the five sets of test data.

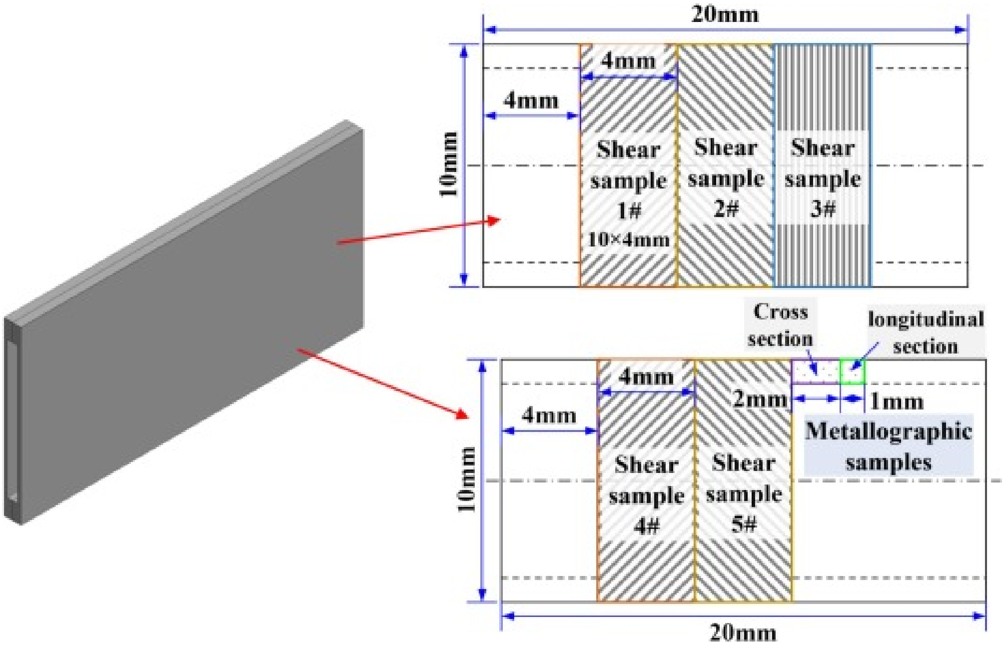

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Fig. 2. Sampling location of metallographic and tensile specimens.

The experimental parameters are shown in Table 1. The comparative experiments were conducted by changing the defocus distances (parameters 1#–5#), single pulse energy (parameters 5#–7#) and welding speed (parameters 5#, 8#, and 9#), respectively.

Table 1. The welding parameters of ultrafast laser.

Test number	Defocus *f* (mm)	Welding speed *v* (mm/s)	Single pulse energy *E* (μJ)
1#	1	1	17.6
2#	0.5	1	17.6
3#	0	1	17.6

Test number	Defocus *f* (mm)	Welding speed *v* (mm/s)	Single pulse energy *E* (μJ)
4#	−0.5	1	17.6
5#	−1	1	17.6
6#	−1	1	15.4
7#	−1	1	20.2
8#	−1	2	17.6
9#	−1	3	17.6

3. Results and discussions

High-repetition-frequency ultrafast laser could heat the areas near the laser spot by heat accumulation [18,19]. During the welding process, ultrafast laser achieves substantial and rapid material melting through heat accumulation effects generated by high-repetition-frequency pulses [11]. Therefore, ultrafast laser at a repetition frequency of 1 MHz is employed in this experiment. A dynamic equilibrium is established between the heat accumulation from ultrafast laser energy into the molten pool and heat dissipation, enabling the formation of stable and sustained molten pool. Consequently, the resultant weld seam (sample 5#) exhibits a uniform macroscopic morphology, characterized by the absence of defects, as shown in Fig. 3.

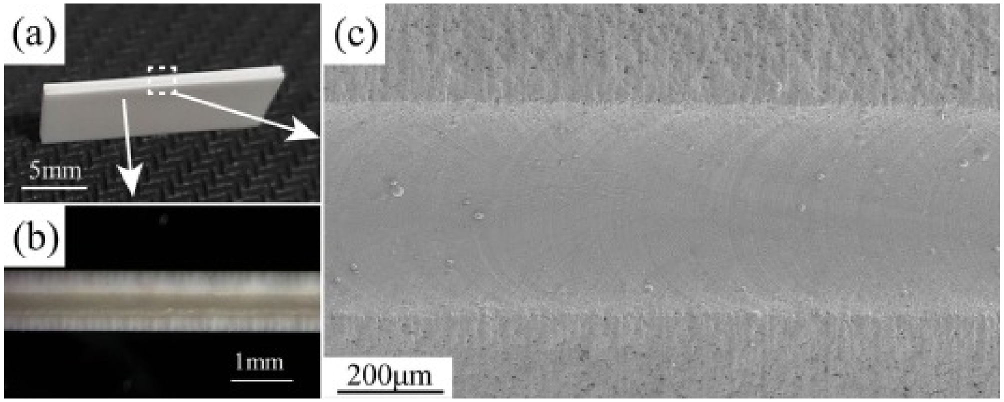

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Fig. 3. Appearance (a) and macro-morphology (b) of the welded sample 5#; (c) SEM of the weld seam.

Fig. 4 depicts the resultant molten tracks formed at different scanning speeds. At 500 mm/s, the minimal melting of the weld seam occurs, with ablation dominating to produce distinct grooves. Reducing the speed to 400 mm/s increases the heat input, intensifying heat accumulation and promoting the formation of discernible weld seam. Further reductions in scanning speed enhance volumetric melting. Progressive increases in melt volume initially generate hump-covered joints. However, the morphology of weld seam gradually transitions towards flattened profile with the decreased speed. As reported by Ai et al. [20] and Ismail et al. [21], hump formation during high-speed scanning (≥500 mm/s) arises from the following mechanism: Elevated scanning speeds coupled with lower heat input generate a narrow and elongated molten pool. Concurrently, the high-energy-density laser induces high-speed plasma plume, propelling the retrograde flow of molten material. The retrograde flow collides with the incoming fluid from the leading edge, resulting in the formation of humps. Rapid solidification resulting from the lower heat input prevents the adequate flow of the humps prior to solidification. As a result, periodic recurrence of the instability produces the hump features [20]. As the scanning speed further decreases, lower scanning speeds shorten the molten pool and stabilize fluid flow, predominantly establishing laminar regimes. Therefore, increased heat input combined with the diminished solidification rates consequently yields the planar surface. In addition, the increased heat input expands the dimensions of molten pool and finally produces smooth weld seam.

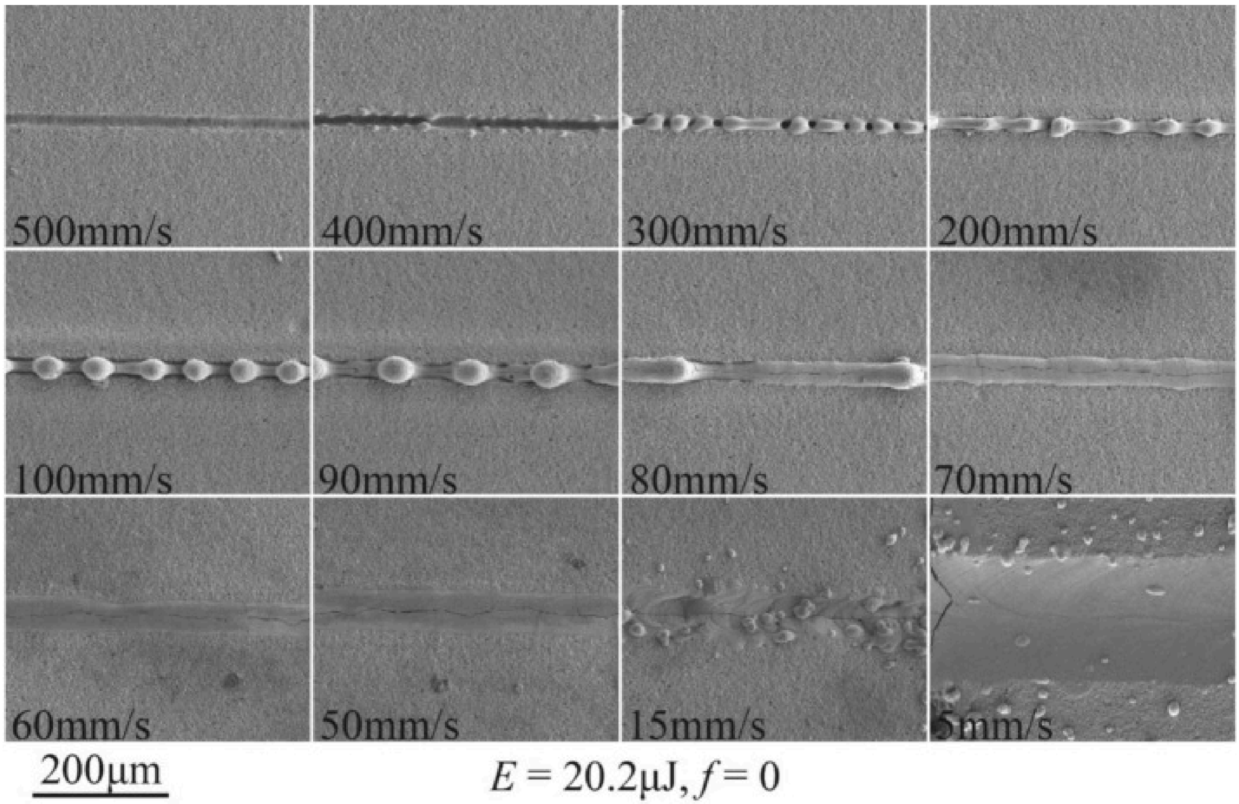

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Fig. 4. The welded morphologies of alumina ceramics sample at different scanning speeds.

The influence of defocus distance on the morphology of weld seam was also investigated. Fig. 5a and b display the surface and cross-sectional morphologies, which reveals the significant variations in the weld depth and width. The characteristic dimensions of weld seam including weld width, depth, depth-to-width ratio (abbreviated as D/W ratio) are illustrated in Fig. 5c and d. When the defocus distance is 1 mm, the D/W ratio of weld seam is relatively low, which is identified as heat conduction mode of laser welding [22]. In the regime, convective heat transfer driven by the Marangoni effect and buoyancy promotes the lateral dissipation of heat from the molten pool center (Fig. 6a), which causes the increased width of weld seam. The above flow direction minimally enhances the weld depth, resulting in the formation of wide and shallow weld seam [22,23]. Furthermore, the weld seam produced at +1 mm defocus exhibits notable pool instability and contains large porosities, which is attributed to the vertical fluctuations of molten material and causes instantaneous variations in the effective defocus distance. These variations periodically elevate laser energy density, intensifying the vaporization within the interaction zone and initiating keyhole formation [20]. However, the formation of keyhole inherently increases the effective positive defocus distance, subsequently reducing the laser energy density. The reduction in laser energy density lowers vaporization-induced recoil pressure below the threshold for keyhole stability. Consequently, the keyhole experiences severe instability and collapse, entrapping vapor and forming porosity. The combination of lower D/W ratio and porosity defects renders these welds highly susceptible to solidification and residual stress cracking, manifesting as severe longitudinal and transverse cracks.

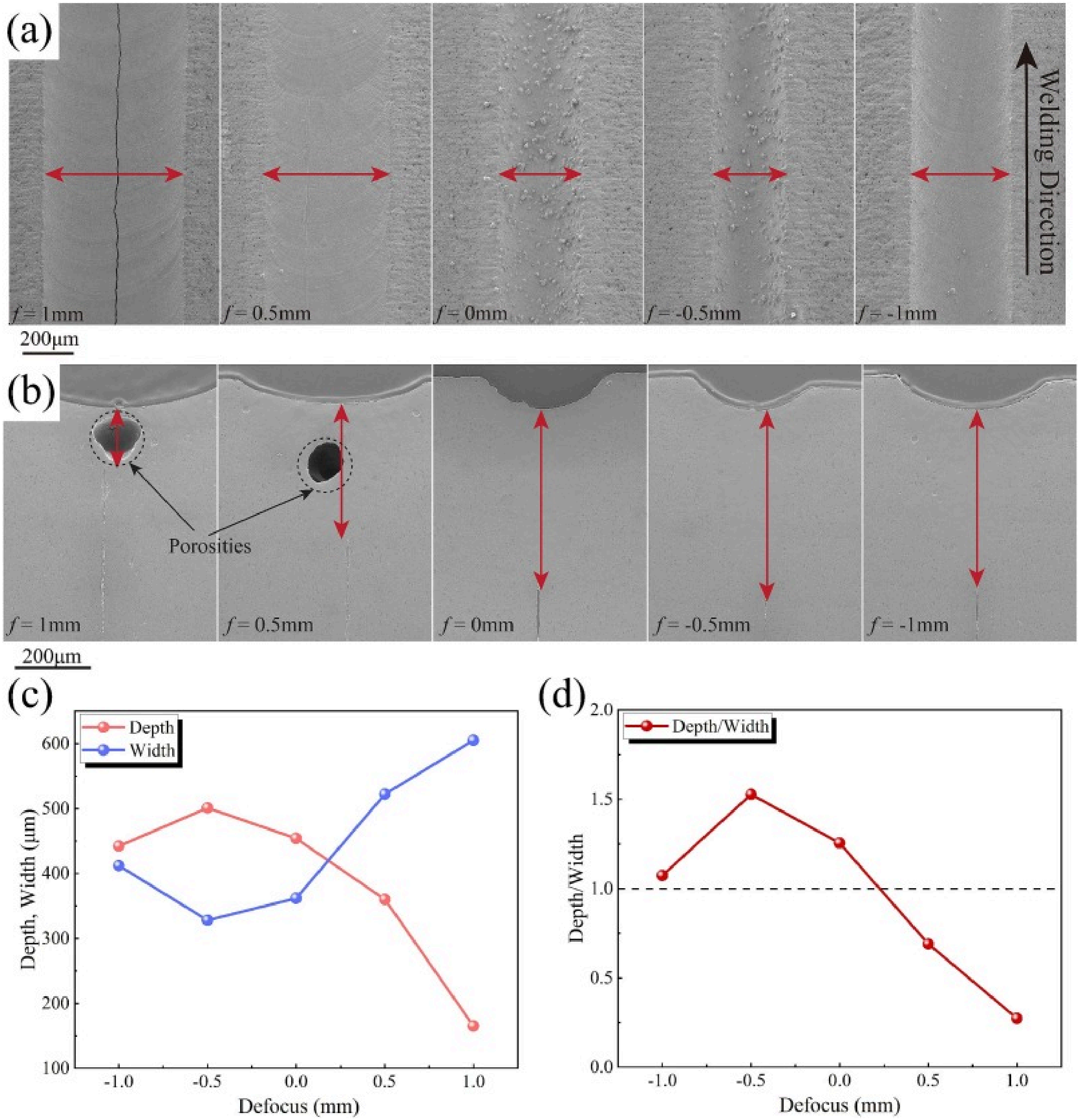

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Fig. 5. Surfaces (a) and cross sections (b) of the welded sample with different defocus distances; the characteristic dimension (c) and D/W ratio (d) of the welded sample with different defocus distances.

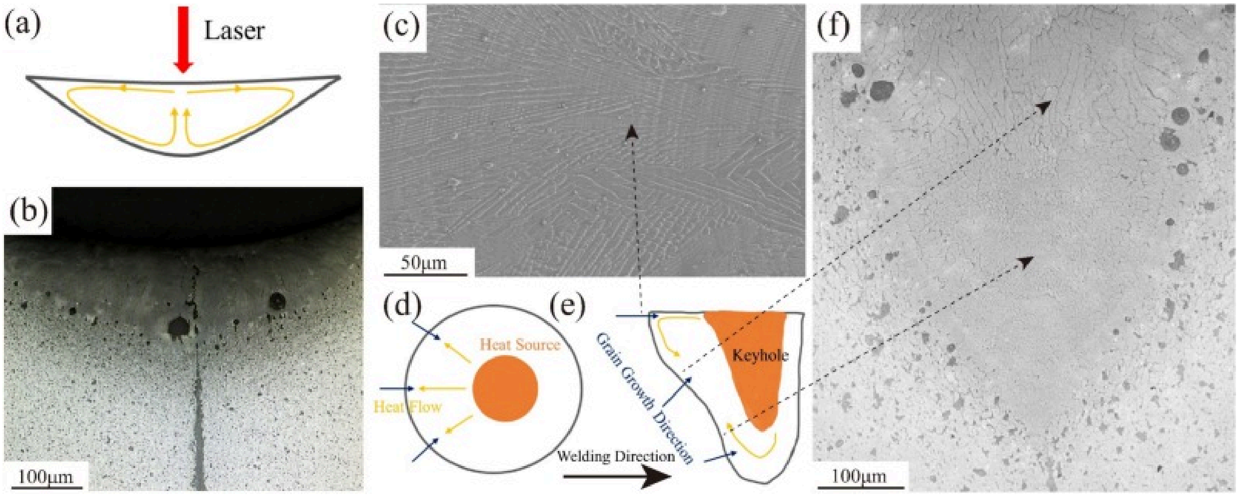

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Fig. 6. (a) Marangoni convection in the heat conduction welding; (b) cross section of the sample 1#; (c) middle area of weld surface in the sample 5#; (d) and (e) convection of melt in laser deep penetration welding; (f) microstructure of the cross section in the sample 5#.

When the defocus distance is set at 0 mm, −0.5 mm, and −1 mm, stable joints with high D/W ratios are obtained. Notably, the maximum D/W ratio (1.53) is achieved at the defocus distance of −0.5 mm. In view of the high repetition rate and short pulse intervals of ultrafast laser, the process approximates a continuous heat source where melting is governed by heat accumulation. Sufficient energy density of ultrafast laser enables to form stable keyhole in the molten pool under the action of metal vapor recoil pressure [20]. According to the high D/W ratio and obvious porosities, it's speculated that the ultrafast laser welding of alumina ceramics presents the typical deep penetration mode.

When the defocus distance increases to +0.5 mm, the case can be considered as a transitional state between conduction welding and deep penetration welding.

Microstructural analysis reveals that the weld seam is predominantly composed of columnar dendrites, consistent with the results in the selective laser melting of alumina [24]. According to the solidification theory, the dendrites grow along the direction of maximum temperature gradient, perpendicular to the melt pool boundaries. During the heat conduction welding for the sample 1# (f=+ 1 mm), the simple convective circulation within the molten pool (as illustrated in Fig. 6a) generates regular temperature distributions near the pool boundaries [22,25]–[26], resulting in the uniform dendritic orientations. Additionally, limited pool depth further restricts the dendritic growth along the depth axis, yielding fewer microstructural layers (Fig. 6b). However, the above dendritic structure offers poor resistance to crack propagation.

Conversely, deep penetration mode by ultrafast laser fosters significantly complex microstructures in the weld seam, which illustrate the different growth directions. The surface morphology of the welded samples at a defocus distance of −1 mm (Fig. 7b~b3) reveals the columnar crystals converging towards the weld centerline. Figs. 6f and 8 demonstrate the microstructure at different locations in the weld seam. The dendrites in the upper-central region exhibit equiaxed morphology due to growth perpendicular to the observation plane (Fig. 8a3). The dendrites in the middle region exhibit an elongated morphology in the cross section (Fig. 8a2). Dendritic structures in the lower-central region (Fig. 8a1) maintains the growth direction consistency with the upper region, but exhibits more refined primary arm spacing. In addition, Fig. 8b also reveals the pronounced epitaxial growth of dendritic structures adjacent to the fusion boundary. The microstructural complexity arises from the intricate thermo-fluid dynamics inherent to the deep penetration welding [[27], [28], [29]].

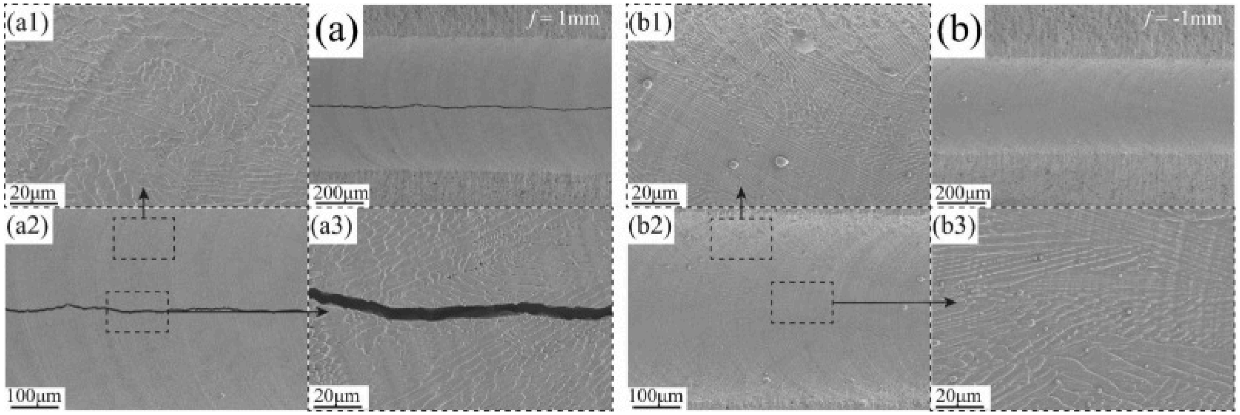

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Fig. 7. (a) sample 1#, surface of the weld seam at f=+ 1 mm; (a2) higher magnification; (a1) edge of the weld seam; (a3) middle area of the weld seam; (b) sample 53, surface of the weld seam at f=−1 mm; (b2) higher magnification; (b1) edge of the weld seam; (b3) middle area of the weld seam.

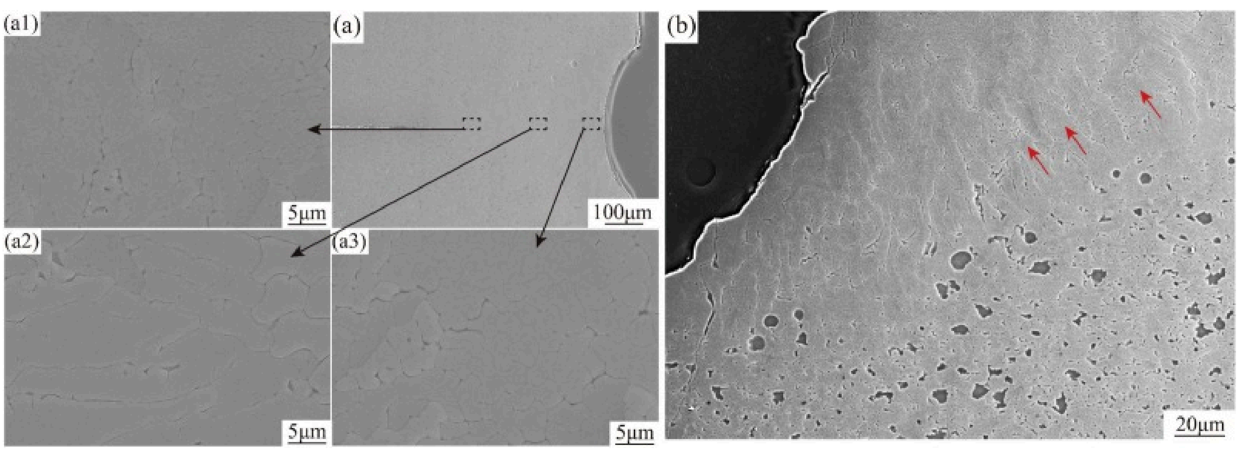

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Fig. 8. (a) sample 5#, cross-section of the weld seam at f=−1 mm; (a1)~(a3) microstructure of different regions of the cross section; (b) microstructure near the fusion boundary.

The fused ceramic at the bottom of the molten pool flows backward, causing the enlarged tail of the molten pool. Concurrently, the molten ceramic on the surface typically flows radially outward from the center of heat source (keyhole) [27]. The flow complexity induces intricate thermal gradients, accounting for the varied dendrite growth orientations observed in Figs. 6c, f, and 8. Moreover, the fast cooling rates of ultrafast laser triggers the refinement of dendritic microstructure in the welded samples [24]. Due to the complex dendritic structure, crack propagation is effectively impeded, significantly reducing the cracking susceptibility.

For the sample 4#, long axial grains parallel to the welding direction are observed in the central region of the welded surface (Fig. 8b3), with curved columnar dendrites distributed on both sides of these axial grains. Notably, axially oriented grains aligned with the welding direction are only present in the weld seam with a depth-to-width (D/W) ratio of approximately 1.53, indicating a higher temperature gradient along the welding direction behind the molten pool relative to other orientations.

Concurrently, severe spatter occurs at defocus distances of 0 mm (Fig. 9) and −0.5 mm (Fig. 10a and b). The spatter exhibits wide spatial distribution. Solidified spatter particles on the weld surface display both spherical and flattened morphologies, consistent with the ejection over varying distances. Furthermore, the spatter is predominantly confined to the weld seam, implying a restricted emission angle. As shown in Fig. 10d and e, spatter formation during the deep penetration welding likely results from the high-speed plasma plume impinging on molten material behind the keyhole [31,32]. The most severe spatter occurs at a defocus distance of 0 mm, potentially due to the higher energy density within the laser-material interaction zone and the consequently increased plasma plume velocity. Spatters could contaminate the weld seam and adjacent regions, therefore it's necessary to suppress the formation of spatters during ultrafast laser welding.

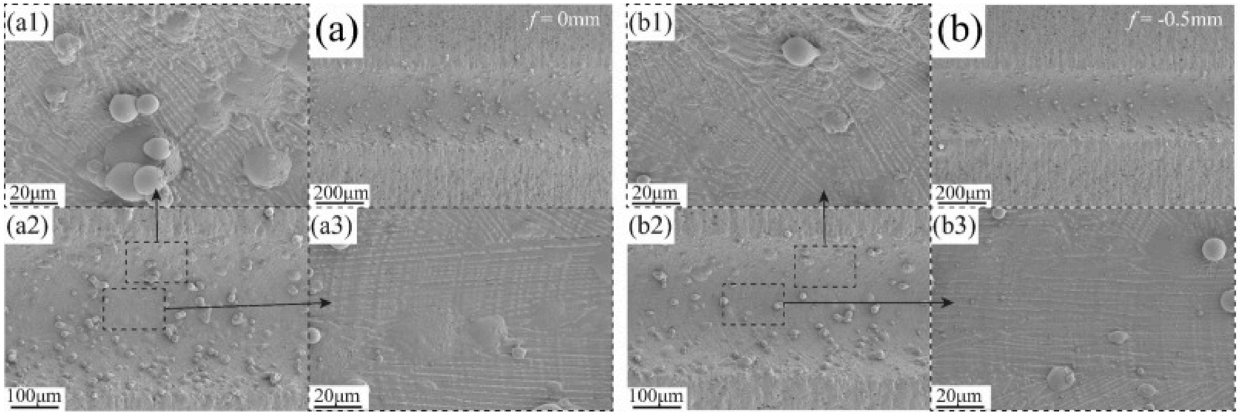

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Fig. 9. (a) Sample 3#, surface of the weld seam at f=0 mm; (a2) higher magnification; (a1) edge of the weld seam; (a3) middle area of the weld seam; (b) sample 4#, surface of the weld seam at f=−0.5 mm; (b2) higher magnification; (b1) edge of the weld seam; (b3) middle area of the weld seam.

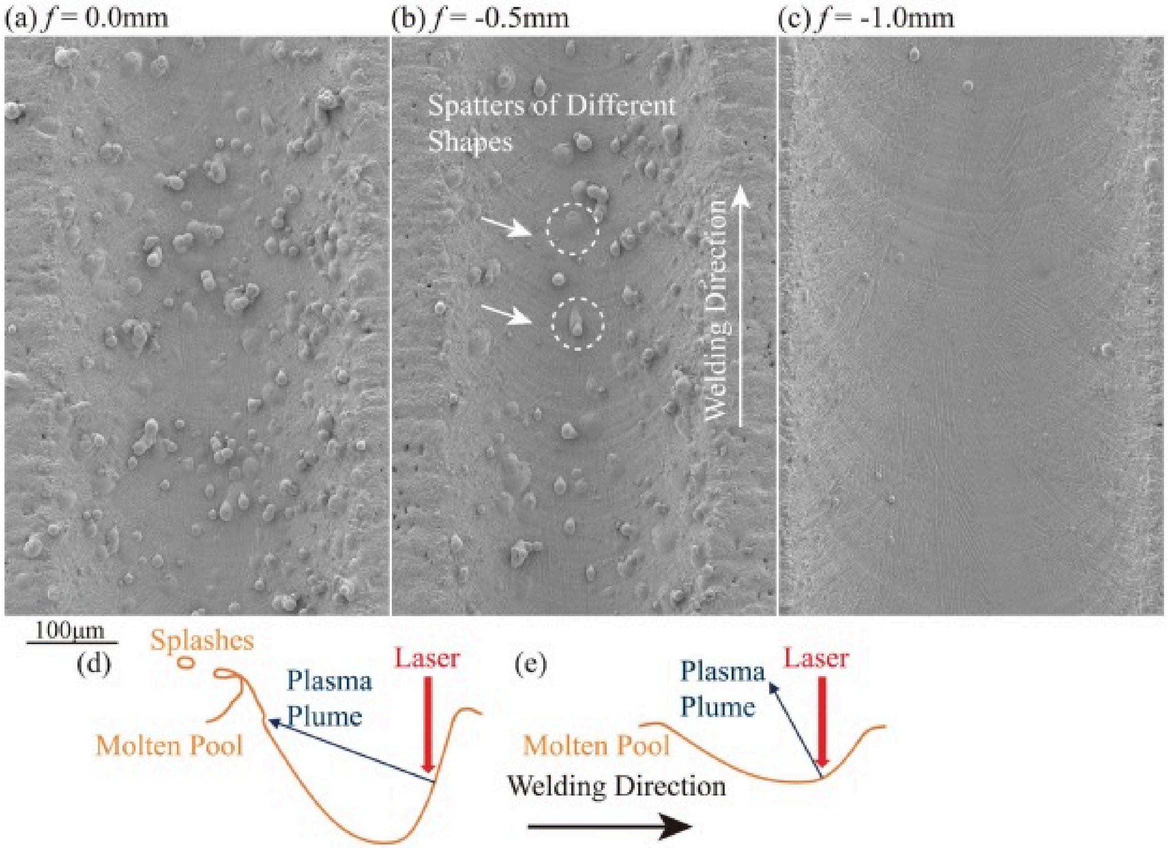

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Fig. 10. Surface of the weld seam in sample 3# (a), 4# (b) and 5#(c); (d)(e) the formation mechanism of spatters under different defocus distance.

At a defocus distance of −1 mm, the welded samples exhibited optimal stability, high D/W ratio, absence of cracks, and minimal surface spatter, yielding the most favorable welded morphology, as shown in Figs. 7b and 10c. Phase characterization was performed by XRD, EDS and TEM. Both the weld seam and BM consist of high-purity α-Al_2O_3 phase, as shown from Fig. 11.

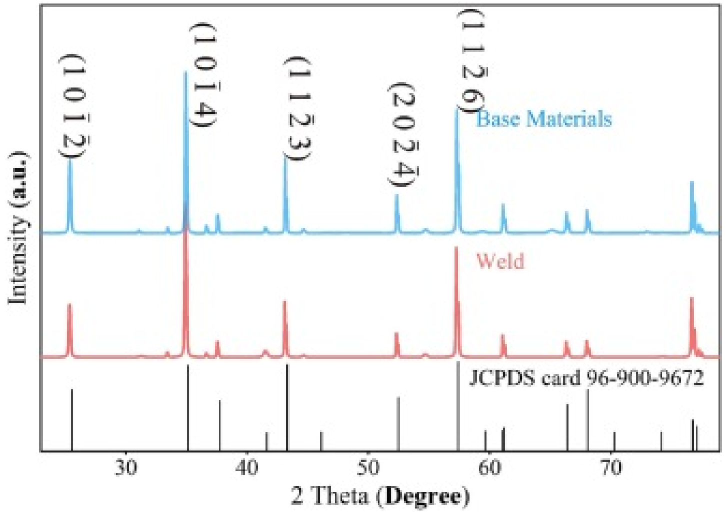

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Fig. 11. XRD of weld seam and BM (sample 5#).

Fig. 12a presents the FIB-prepared TEM lamella, revealing the microstructure of weld seam. The elemental mapping results (Fig. 12b) distinguishes the dendrites (bright contrast) from the interdendritic regions (dark contrast). EDS analysis indicates that the dendrites are primarily composed of Al and O, while the interdendritic regions contain Si, O, and minor amounts of Ca and Mg elements.

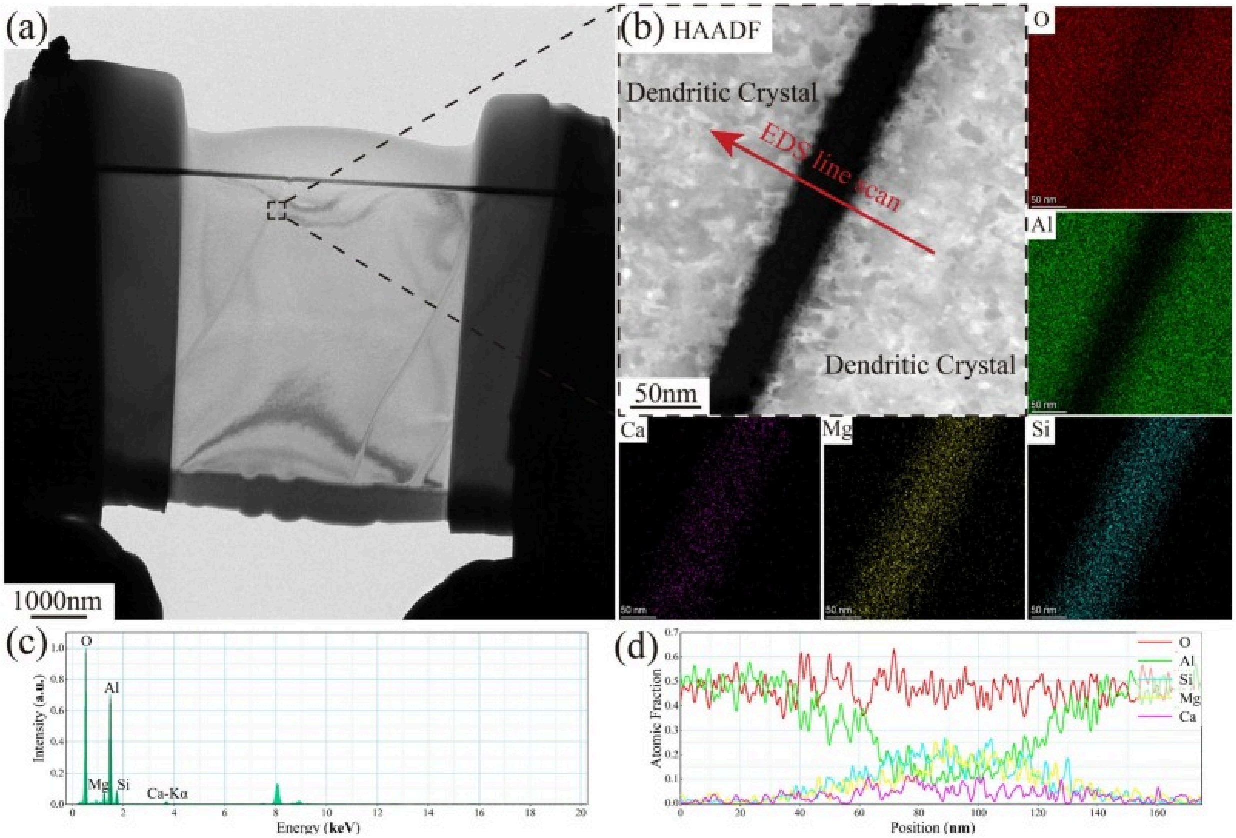

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Fig. 12. EDS analysis of microstructure in weld seam (sample 5#): (a) dendritic-glass-phase-dendritic sheet cut from the weld; (b) distribution of major elements; (c) content of major elements; (d) distribution of elements along the scanning path in (b).

Selected area electron diffraction (SAED) patterns were analyzed based on the corresponding elemental results. High-resolution TEM (HRTEM) images and SAED patterns in the regions 1, 2, and 3 are shown in Fig. 13b, c, and d, respectively. The dendritic regions (Fig. 12b and c) exhibit the sharp and well-defined SAED spots, confirming the high-purity α-Al_2O_3 phase. In contrast, the interdendritic regions (Fig. 13d) display diffuse ring patterns, indicating the presence of the glass phase.

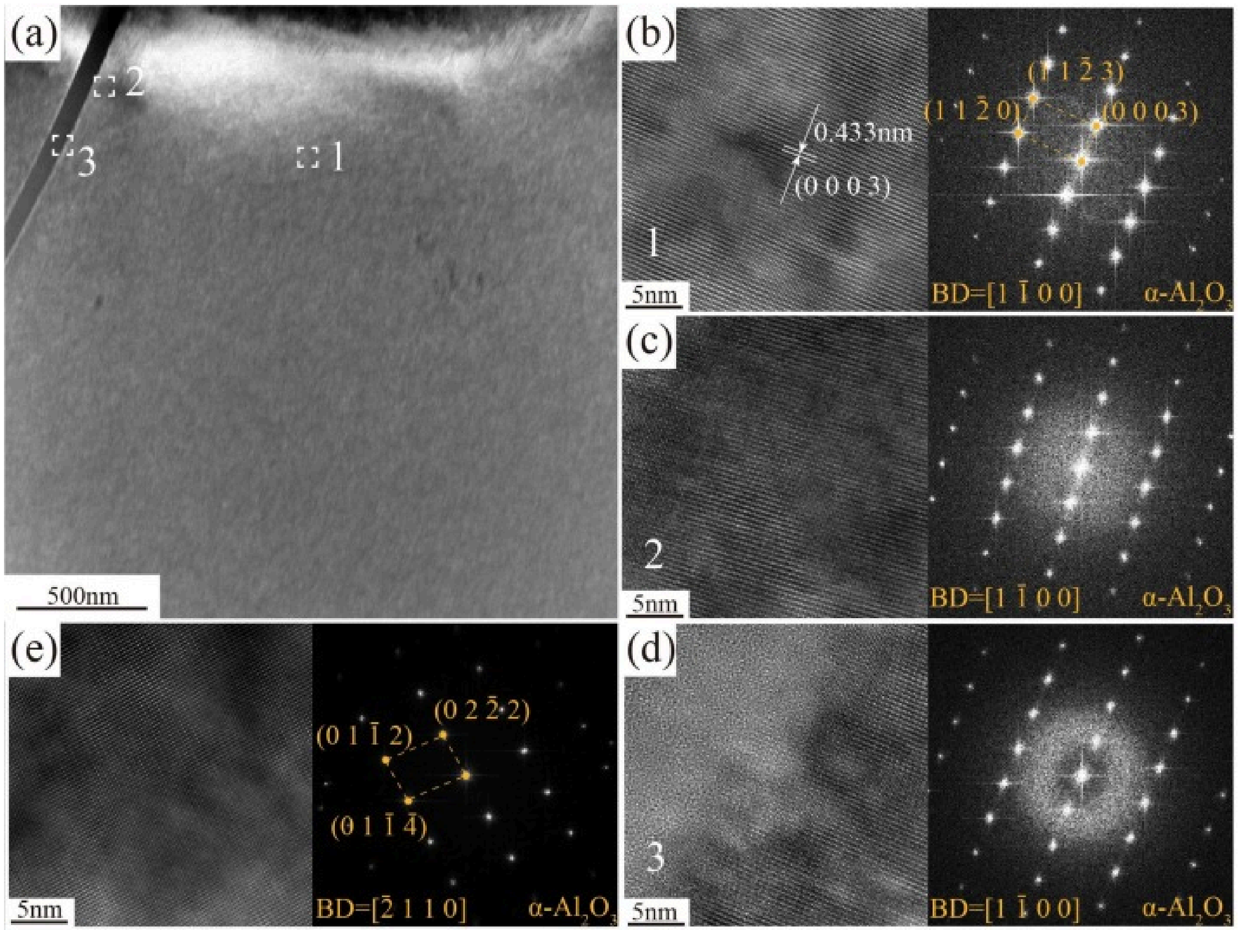

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Fig. 13. TEM analysis of microstructure in welds (sample 5#): (a) dendritic-glass-phase-dendritic sheet cut from the weld seam; (b) HRTEM and SAED patterns of point 1; (c) HRTEM and SAED patterns of point 2; (d) HRTEM and SAED patterns of point 3; (e) HRTEM and SAED patterns of point 1.

The weld seam is composed of predominantly α-Al_2O_3 and minor glassy phase, which is similar to the microstructure of BM. It's consistent with the results from the prepared alumina ceramics by laser directed energy deposition [33]. The grain orientations of weld seam along the different axes are presented in Fig. 14b, c, and d, respectively. The corresponding pole figures (PFs) and inverse pole figures (IPFs) of the BM and weld seam are shown in Fig. 14e and f. As can be seen from Fig. 14e1 and f1, BM exhibits a distinct fiber texture in the X-axis direction, with grains preferentially oriented near the [0001] direction. No significant texture is observed in other directions. The preferential orientation aligns with the crystallography of α-Al_2O_3 phase, where the (0001) close-packed plane promotes the growth along [0001].

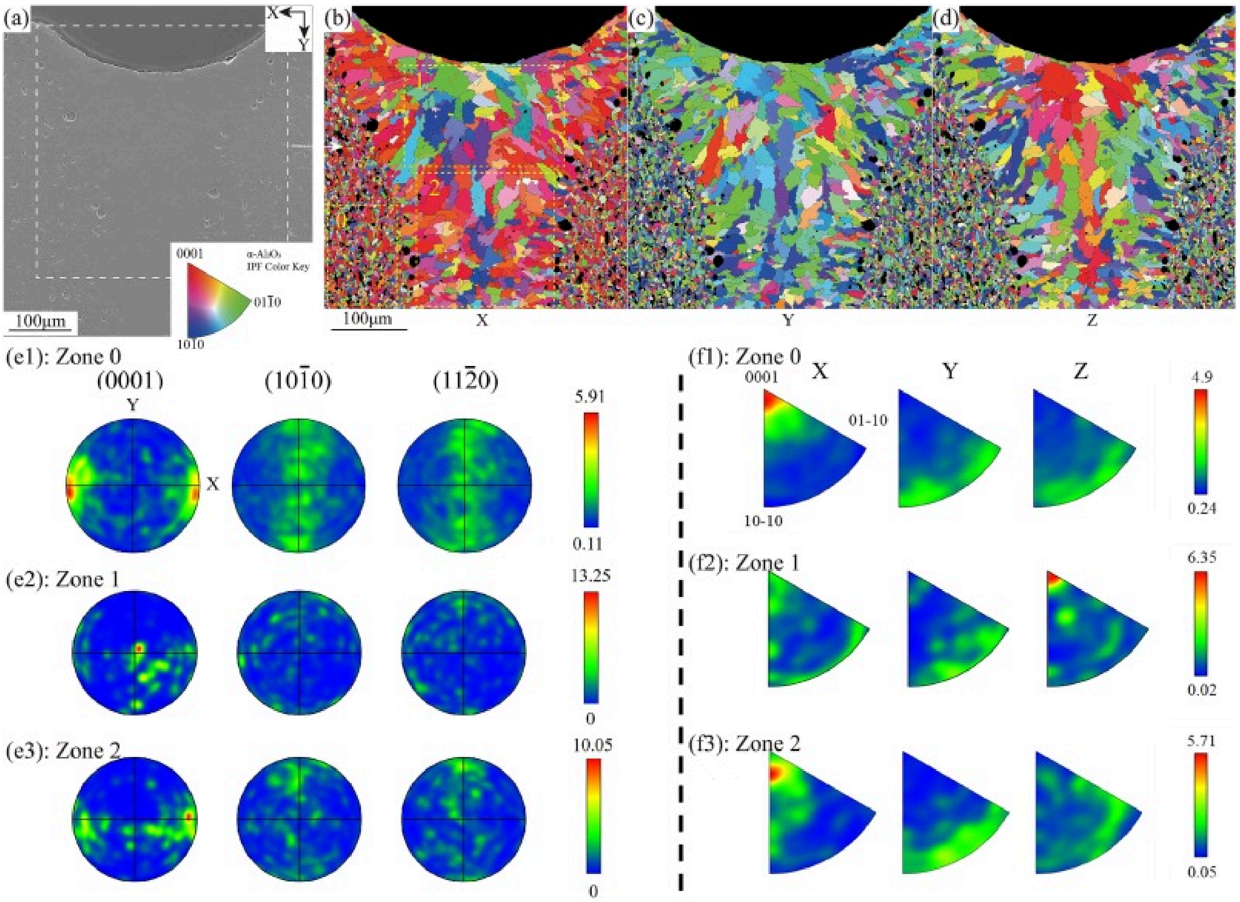

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Fig. 14. EBSD analysis of weld seam (sample 5#): (a) cross section; (b)~(d) grain orientation distribution; (e1)~(e3) PFs of different zones in (b); (f1)-(f3) IPFs of different zones in (b).

It can be seen from Fig. 14e2 and f2 that the grain orientation along *Z*-axis direction in the middle region of the weld seam is biased [0001], which is different from the grain orientation of BM. It likely arises from solidification in the region, less constrained by the BM, allows grains to grow preferentially along the maximum heat flow direction (Z-axis), corresponding to [0001]. Conversely, the narrow lower-middle weld region exhibits a higher proportion of epitaxial grains originating from the BM. Consequently, grain orientation along the X-axis is influenced by the BM, also favoring [0001]. However, the bending-induced maximum temperature gradient causes a slight deviation from the exact [0001] orientation.

It's observed that both of weld depth and width increase with the increased single pulse energy in Fig. 15a–c. As shown in Fig. 15d, the dendritic morphology primarily depends on the three solidification parameters at the growth front: solute concentration (C_0), temperature gradient (G), and crystallization rate (R). The increase in single pulse energy causes the significant reduction of G, thereby intensifying the secondary dendritic growth.

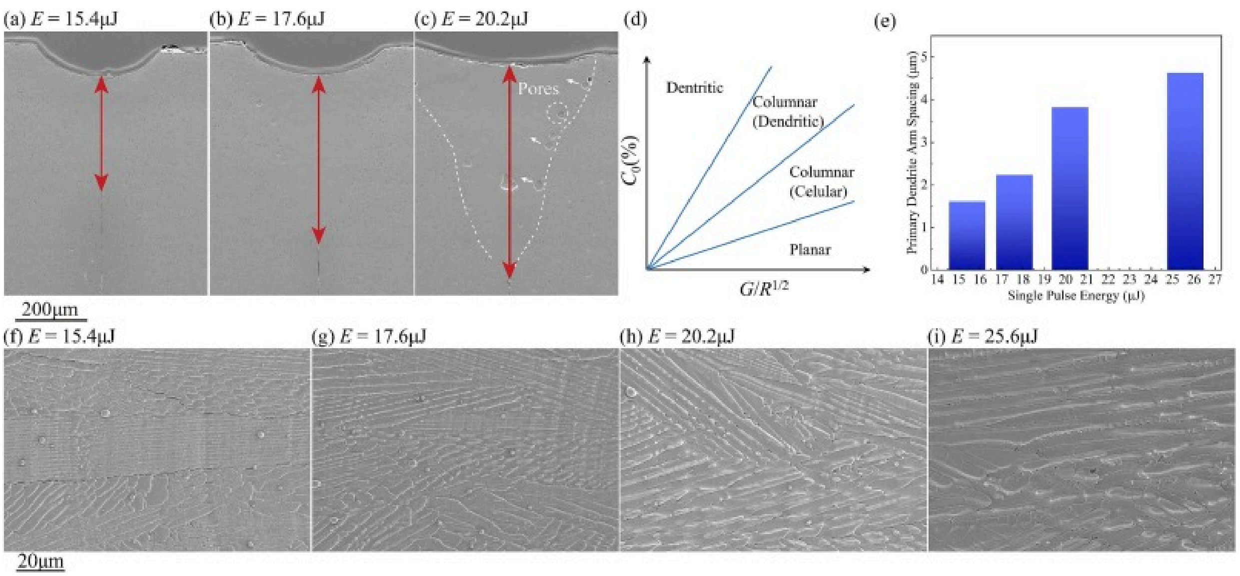

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Fig. 15. (a), (b) and (c) Cross section at different single pulse energies (sample 6#, 5# and 7#); (d) dendrite type versus solute concentration C_0, temperature gradient G and crystallization rate R; (e) effect of single pulse energy on primary dendrite spacing along

the centerline region on the weld surface; (f)~(i) corresponding dendrites along the centerline region on the weld surface.

As evidenced in Fig. 15e, the dendritic arm spacing is fundamentally controlled by the cooling rate (expressed as $G \times R$). The enhancement of single pulse energy reduces the $G \times R$ value, resulting in the progressive coarsening of the primary dendritic spacing. Insufficient single pulse energy will lead to insufficient melting, preventing the components from achieving reliable joining. However, excessive single-pulse energy increases melt pool volume, making it difficult for vaporization-induced recoil pressure to maintain the keyhole stability. Consequently, the probability of keyhole collapse increases, leading to large porosity formation within the joints, as validated in Fig. 15c. The small porosities observed near the fusion line primarily originate from the pre-existing pores within BM [15]. The work points out that such porosity is exclusively located near the fusion boundary, with minimal upward migration observed. It's therefore hypothesized that the lower temperature near the fusion boundary, combined with higher surface energy and poor wettability with the solid phase, collectively prevents the pores at the solid-liquid interface from detaching. As shown in Fig. 15a, b, and c, the size of these small pores near the fusion boundary increases with higher single-pulse energy. These enlarged pores are positioned at a distance from the fusion boundary, tending towards the weld interior. The phenomenon may be attributed to the elevated temperature at the molten pool boundary, which reduces the surface energy and improves the wettability. It enables the small pores, initially adhering to the solid-liquid interface, to detach and coalesce into larger pores.

As shown in Fig. 16a–c, the weld depth and width decrease gradually with the increased welding speed. According to the crystallization speed equation ($R = v \cdot \cos\alpha$), higher welding speeds increase the crystallization rate and temperature gradient. Consequently, the cooling rate increases significantly, resulting in the reduce of primary dendrite arm spacing (Fig. 16d). Under welding speeds of 2–3 mm/s, severe transverse cracks and micro-cracks distributed along the interdendritic regions are presence in the weld seams (Fig. 17). These transverse cracks are primarily attributed to two factors: (1) the increased temperature gradient, which intensifies localized and irreversible plastic deformation, and (2) the elevated cooling rate, which shortens the time available for stress release via plastic deformation during the cooling process. The combined effect substantially increases residual tensile stress within the joint. In addition, the alumina ceramics used in this work contain 95–96 wt% α-Al_2O_3, with the remainder consisting predominantly of glass phase formed by sintering SiO_2, MgO, and CaO. The disparity of solidus temperature between the glass phase and α-Al_2O_3 creates the interfacial conditions favorable for the initiation of solidification crack. While α-Al_2O_3 melts at 2054 °C, the SiO_2-dominated glass phase transitions to a viscous state at lower temperatures (1600–1700 °C). The substantial melting point difference widens the solidification temperature range. Thus, these micro-cracks are classified as solidification cracks. During solidification, high-purity α-Al_2O_3 (characterized by high elastic modulus and low plasticity) crystallizes preferentially, leaving a minimal residual viscous glass phase in the intergranular regions of the solidified structure. Qiu et al. pointed out that a significant volume shrinkage (33%) would appear when the ceramic melt turns into solid [34]. As a result, insufficient stress relaxation capacity increases the susceptibility of the solid phase to tensile tearing under the rapid cooling and high strain rates, causing intergranular crack [35]. These solidification cracks occur both at the weld surface (Fig. 17c, d) and within subsurface regions (Fig. 17f, g).

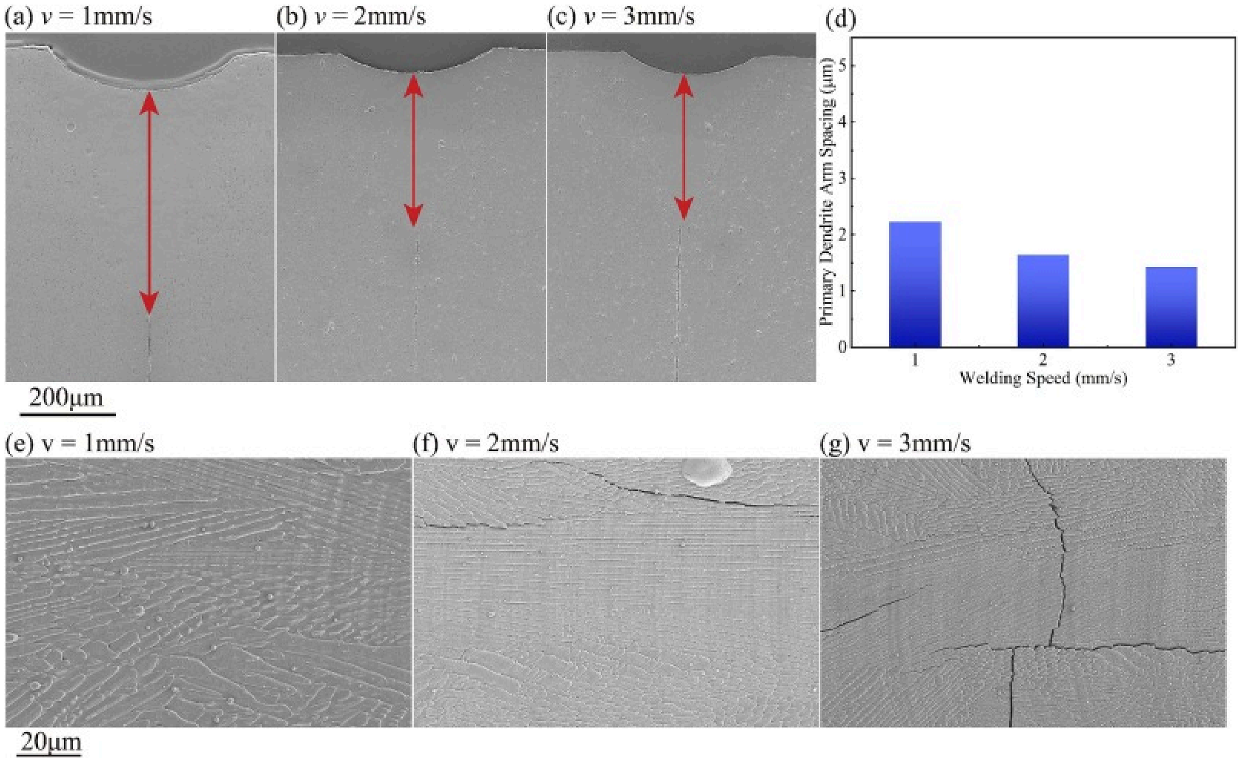

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Fig. 16. (a), (b) and (c) Cross section at different welding speeds (sample 5#, 8# and 9#); (d) effect of welding speed on primary dendrite spacing along the centerline region on the weld surface; (e)~(g) corresponding dendrites along the centerline region on the weld surface.

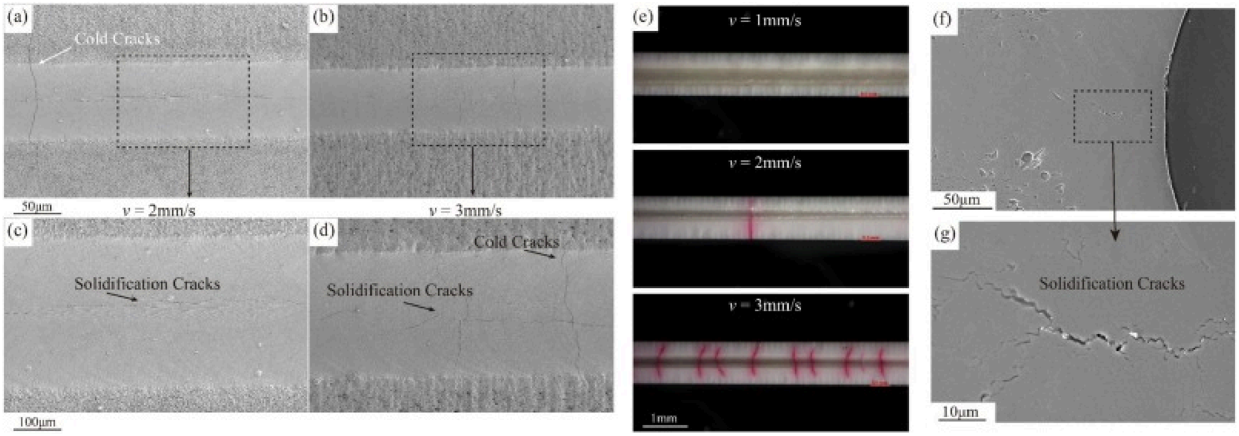

Fig. 17. (a)(c) sample 8#, surface of the joint at v=2mm/s; (b)(d) sample 9#, surface of the weld seam at v=3mm/s; (e) comparison of weld seam with different welding speeds; (f)(g) internal solidification cracks.

Fig. 18 shows the metallographic photos near the fusion line and heat affected zone (HAZ). The joints fabricated with single-pulse energy of 15.4μJ (sample 6#) and scanning speeds of 2mm/s or 3mm/s (sample 8# and 9#) exhibit a higher density of defects within the cross-section of HAZ, including significant fragmentation and observable cracking. The phenomenon is likely attributable to the thermal softening of intergranular glassy phases. In their weakened state, these phases experience excessive tensile stresses induced by solidification contraction of the molten pool, ultimately the formation of initiating crack. Increasing the single-pulse energy or decreasing the scanning speed reduces the temperature gradient, which diminishes the non-uniform deformation and associated tensile stresses during cooling. Concurrently, the resultant lower cooling rate and strain rate significantly mitigate the cracking susceptibility.

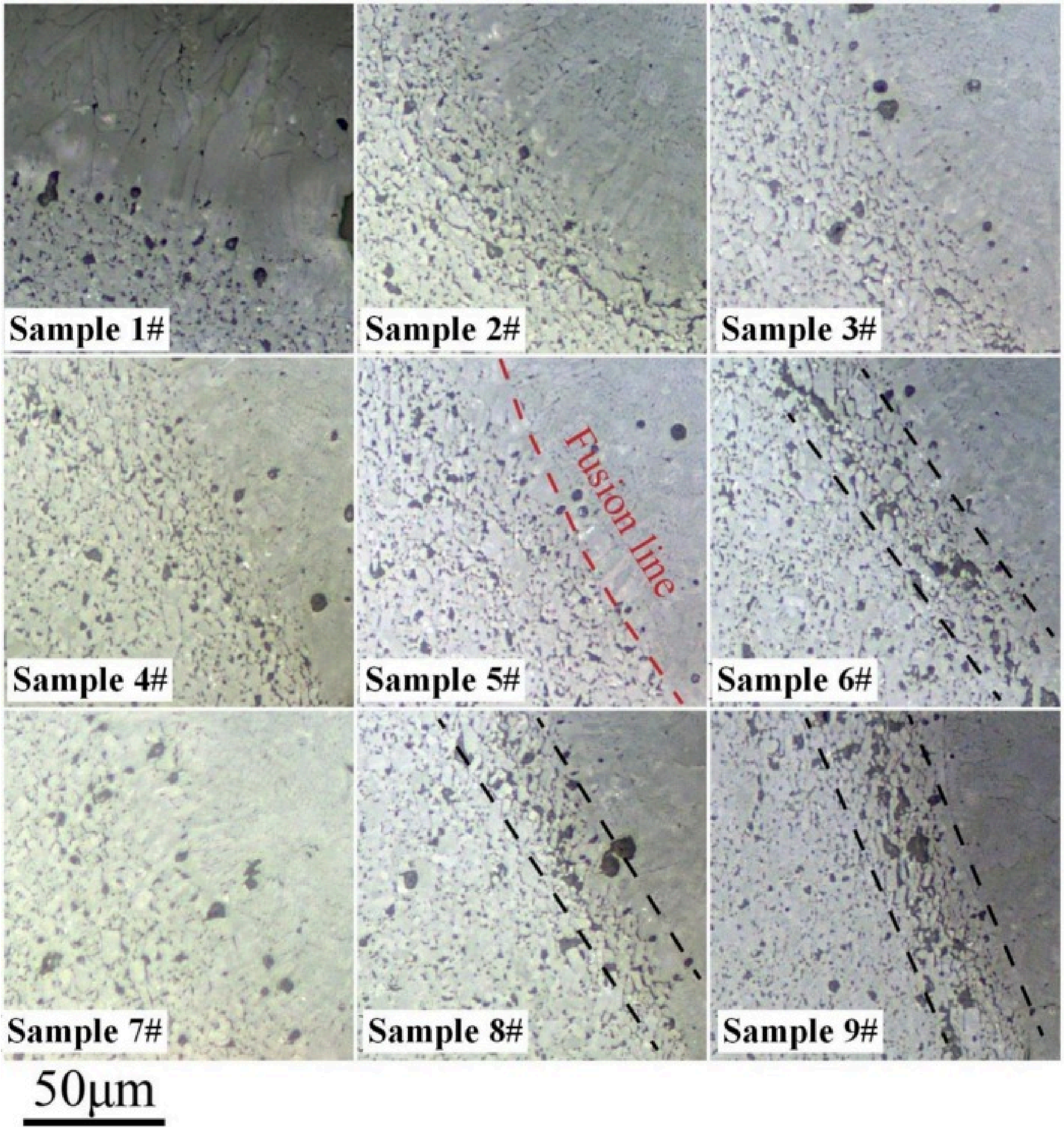

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Fig. 18. Fusion lines and HAZ exhibited in the joints with different welding parameters.

The weld seam and BM exhibit the identical phase compositions. However, the weld seam demonstrates significantly larger grain sizes compared to BM. Furthermore, the weld seam predominantly contains columnar grains, presenting a morphological distinction from the equiaxed grains of the BM. The weld zone can be regarded as a distinct alumina ceramic material with elevated purity and density levels. Critical microstructural parameters including grain size, grain morphology, grain boundary content, and pore volume fraction collectively influence mechanical properties of alumina ceramic, resulting in a complex structure-property relationship. However, for high-purity alumina ceramics, the inability to undergo plastic deformation at ambient temperature renders variations in grain size within a specific range less influential on mechanical properties [36]. The reasons for the differences in mechanical properties between the weld seam and BM may lie more in aspects such as the morphology of grains and pore content.

Micro-Vickers hardness measurements were conducted on the cross section of the joint under the testing parameters: 1 kgf indentation load with 10 s dwell time. Hardness variations across distinct regions of the joint are explicitly illustrated in Fig. 19a and b. The substantial porosity of the BM significantly compromises the microhardness, resulting in an approximate hardness value of 1300 HV. The microhardness near the fusion line is comparable to the BM, while the weld seam exhibits significantly higher microhardness than the BM. Compared to the BM, the weld zone demonstrates higher relative density. However, due to the microstructure comprising dendrites and interdendritic glassy phases, the microhardness of the joint still lower than that of pure α-Al_2O_3 single crystals (1940 HV measured parallel to the c-axis [37]). The microhardness at the center of the joint (point 3) under different parameters is compared in Fig. 19c. It can be observed that the effect of welding parameters on the joint microhardness is not significant. The microhardness of the joint generally ranges from 1500 HV to 1600 HV.

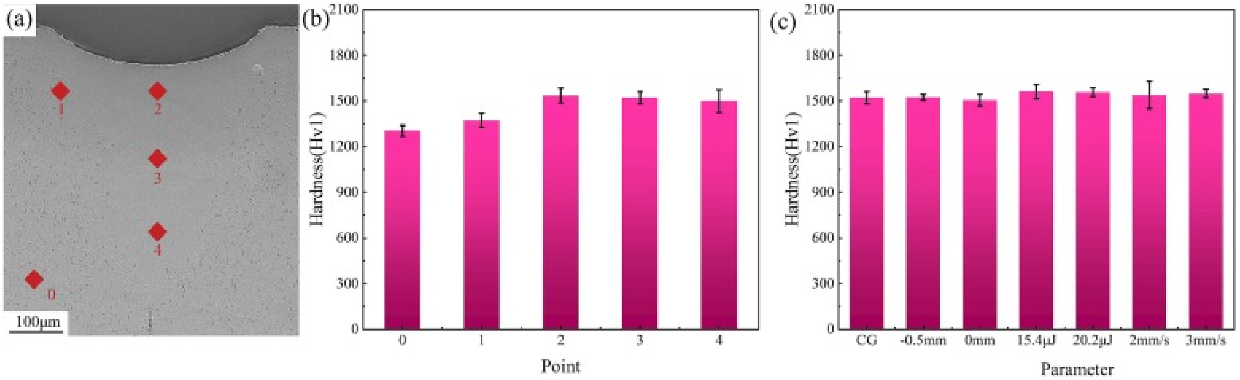

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Fig. 19. (a) Position of the pickup points for the data in (b); (b) the microhardness of points in (a); (c) influence of parameters on microhardness.

The shear strength test of the welded sample was conducted. The fixture used for the shear strength testing and the method of applying the shear force are shown in Fig. 20a and the tested results are presented in Fig. 20b. The results indicate that the joints with high width-to-depth (W/D) ratio, produced within a defocus distance range of −1 to 0 mm, exhibit superior shear strength. In contrast, a pronounced decrease in shear strength is observed when the defocus distance is increased to the 0.5–1 mm range. The effect of single-pulse energy on shear strength is shown in Fig. 20c. The maximum shear strength of 90.9 MPa is achieved at a single-pulse energy of 20.2 μJ. A reduction in single pulse energy results in the diminished shear strength, with a particularly pronounced decrease observed at 15.4 μJ. Overall, the joints with high depth-to-width (D/W) ratio, fabricated by higher single-pulse energy levels in combination with negative defocus distances, demonstrate superior shear strength, typically ranging from 80 MPa to 90 MPa, with certain specimens exceeding 90 MPa.

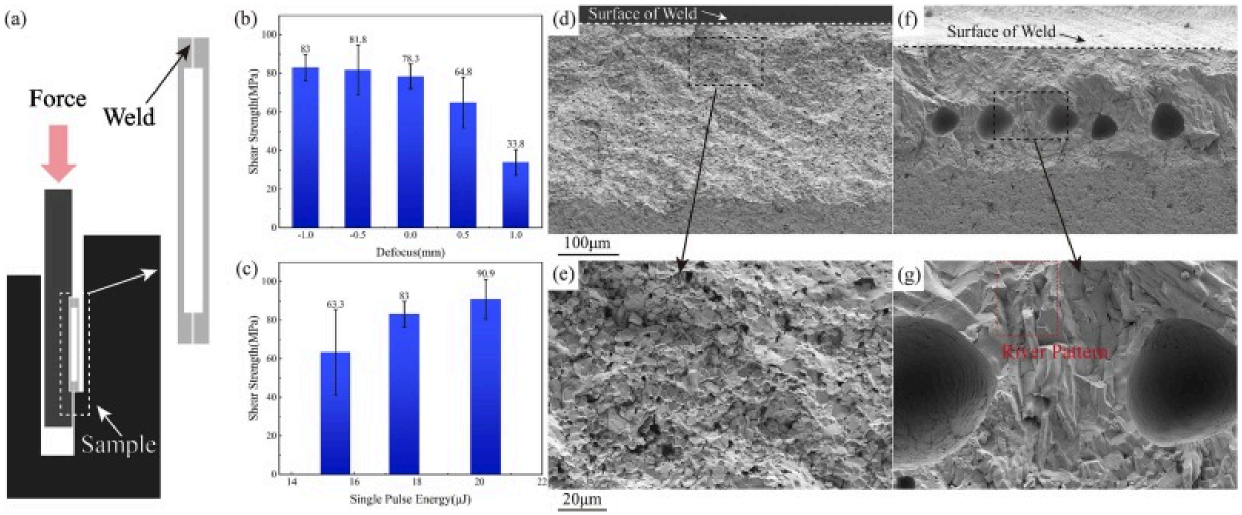

Fig. 20. (a) Schematic diagram of shear mechanical testing; (b) effect of defocus on shear strength (sample 1#–5#); (c) effect of single pulse energy on shear strength (sample 6#, 5# and 7#); (d)(e) shear test fracture for sample 6#; (f)(g) shear test fracture for sample 1#.

Fractographic analysis of the shear-tested specimens in Fig. 20b reveals that the joint with high D/W ratio fracture near the fusion line. The fractured surface in Fig. 20d exhibits a obvious rock sugar-like morphology, demonstrating the typical intergranular brittle fracture characteristics. As seen from the EBSD images, it can be seen that there is a sudden transition in grain size adjacent to the fusion boundary. The severe grain size mismatch is susceptible to stress concentration under the mechanical loading, thereby promoting crack initiation in this region. The cracks observed near the fusion line of the joints (single-pulse energy: 15.4 μJ) is likely responsible for the significant reduction in shear strength. As shown in Fig. 20d and g, the defocus distance of 1 mm results in the formation of micro-cracks within the weld interior. It is attributed to the prevalence of longitudinal cracks and severe porosity at the defocus distance of 1 mm, which substantially degrade the weld integrity. Furthermore, the fractured surface exhibits the features of river pattern, confirming the brittle transgranular fracture mode.

Travessa et al. utilized Cu interlayer to join alumina ceramics and stainless steel at 700–900 °C by diffusion welding, achieving the highest shear strength of 65 MPa [38]. Similarly, Li et al. achieved the brazed joining of high-purity alumina and oxygen-free copper with Ag-Cu-Ti filler, attaining a peak shear strength of approximately 90 MPa [39]. In contrast, the alumina ceramics is joined by ultrafast laser at room temperature within tens of seconds, and the obtained joints present the optimal shear strength of up to 90.9 MPa. These results highlight the advantages of ultrafast laser welding for achieving high-quality alumina joints with significantly reduced processing time.

4. Conclusions

In summary, the welded samples are obtained by ultrafast laser welding, consisting of high-purity α-Al_2O_3 and glass phases. The welded samples exhibit significantly larger columnar dendrites. The primary defects in the welded samples include cold cracks, solidification cracks, porosity, and spatter. The presence of porosity and cracks resulted in a significant decrease in mechanical properties of the welded sample. Optimized process parameters (−1 mm defocus distance, 17.6 μJ single pulse energy, 1 mm/s welding speed) yielded the high-quality welded samples characterized by free of cracks, flat surface, minimal porosities and spatter. The sound joint (sample 5#) exhibits the highest shear strengths, approximately 80–90 MPa. This work provides the strong evidence for the feasibility of high-quality melting welding of alumina ceramics by ultrafast laser.

CRediT authorship contribution statement

Jian Zhao: Writing – original draft, Investigation, Funding acquisition, Data curation. **Qingui Fan:** Writing – original draft, Methodology, Data curation. **Haikun Liu:** Methodology, Investigation. **Yijie Pu:** Software, Formal analysis. **Fangzhou Li:** Validation, Formal analysis. **Wenhui Xue:** Data curation. **Zixuan Zhang:** Methodology. **Xiaopeng Li:** Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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